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Integrating Spatial and Temporal Features for Object Motion Detection using CNN–BiLSTM Architecture

1st Divya S B*
Department of Computer Applications,
Mangalam College of Engineering,
Kottayam,Kerala.
divyabhadran@gmail.com

2nd Anjaly Sajeev
Department of Computer Applications,
Mangalam College of Engineering,
Kottayam,Kerala.
anjali.sajeev@mangalam.in

3rd Sreelakshmi
Department of Computer Applications,
Mangalam College of Engineering,
Kottayam,Kerala.
sreelakshminair1818@gmail.com

4th Neha Mary John
Department of Computer Applications,
Mangalam College of Engineering,
Kottayam,Kerala.
nehamary996@gmail.com

5th Rosemary B
Department of Computer Applications,
Mangalam College of Engineering,
Kottayam,Kerala.
rosemary200327@gmail.com

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Abstract- **Object motion detection plays a vital role in video analysis applications such as surveillance, autonomous navigation, and human–computer interaction. This study presents an object motion detection approach that integrates Convolutional Neural Networks (CNN) with Bidirectional Long Short-Term Memory (Bi-LSTM) networks to effectively capture both spatial and temporal information from video sequences. The CNN component is employed to extract discriminative spatial features from consecutive frames, while the Bi-LSTM model learns bidirectional temporal dependencies to accurately identify motion patterns over time. The combined architecture enhances robustness against background noise, illumination changes, and complex object movements. Experimental evaluations conducted on benchmark video datasets demonstrate that the proposed CNN-Bi-LSTM framework achieves improved motion detection accuracy and temporal consistency compared to traditional and single-model approaches. The results confirm the effectiveness of combining deep spatial representations with sequential learning for reliable object motion detection in dynamic environments.**

Keywords: CNN,BI-LSTM, Object Motion, Human Computer Interaction, Single-Model.

1. INTRODUCTION

Object motion detection [1] is a fundamental task in computer vision with applications in video surveillance, autonomous systems, activity recognition, and traffic monitoring. Accurate detection of moving objects enables intelligent systems to understand dynamic environments and respond effectively to changes over time. Traditional motion detection methods often rely on handcrafted features and frame differencing, which are sensitive to noise, illumination variations, and complex backgrounds.

With the rapid growth of video data, deep learning approaches have become increasingly popular for extracting meaningful representations from visual content. Convolutional Neural Networks (CNNs) [2] are particularly effective in learning spatial features such as object shape, texture, and edges. However, CNN-based models alone are limited in capturing temporal dependencies essential for understanding motion dynamics across video frames.

Temporal modeling plays a critical role in distinguishing true object motion from background fluctuations. Recurrent neural networks, especially Long Short-Term Memory (LSTM) [3] networks, have demonstrated strong capabilities in modeling sequential data. By learning temporal relationships, LSTM-based approaches enhance motion continuity and reduce false detections caused by sudden environmental changes.

Bidirectional LSTM (BiLSTM) [4] networks further extend temporal learning by processing sequences in both forward and backward directions. This bidirectional context allows the model to better capture motion evolution and dependencies across time, improving detection

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accuracy in complex scenes with occlusions or irregular movements.

Motivated by these advancements, this study proposes a hybrid CNN–BiLSTM architecture that integrates spatial and temporal feature learning for robust object motion detection. The proposed approach aims to improve detection accuracy, temporal consistency, and resilience to challenging video conditions.

2. Related Work

Early object motion detection techniques [5] were based on background subtraction and optical flow analysis. While computationally efficient, these methods struggled with dynamic backgrounds, camera motion, and illumination changes. Their reliance on predefined rules limited adaptability to complex real-world scenarios.

The introduction of machine learning methods [6] enabled more flexible motion detection by learning patterns from data. However, traditional classifiers required handcrafted features, which often failed to generalize across diverse environments. Feature extraction remained a significant challenge in these approaches.

CNN-based models marked a major shift by automatically learning spatial features from video frames. Several studies demonstrated that CNNs outperform traditional methods [7] in object detection and segmentation tasks. Despite their success, CNNs primarily focus on spatial information and lack temporal awareness when applied independently.

To address temporal modelling [8], researchers incorporated recurrent neural networks, particularly LSTMs, into video analysis tasks. CNN-LSTM hybrids became popular for action recognition and event detection, enabling models to learn frame-to-frame dependencies. However, unidirectional LSTMs sometimes failed to capture complete motion context.

Recent studies have explored BiLSTM-based architectures to enhance temporal understanding. By integrating CNNs with BiLSTM networks, researchers achieved improved performance in motion-sensitive tasks. Nevertheless, there remains a need for optimized architectures that effectively balance spatial feature richness and temporal continuity, which this research seeks to address.

3. Methodology Proposed

The proposed methodology follows a structured pipeline designed to integrate spatial and temporal information for effective object motion detection. The process begins with video frame extraction [9], where continuous video streams are converted into ordered frame sequences suitable for deep learning analysis.

In the spatial feature extraction stage, a CNN is employed to learn discriminative visual features from each frame. Convolutional and pooling layers capture essential information related to object boundaries, textures [10], and movement cues while reducing spatial dimensionality. This stage ensures robust representation of visual content.

The extracted spatial features are then fed into a BiLSTM network to model temporal dependencies. Unlike unidirectional LSTMs, the BiLSTM processes sequences in both temporal directions, enabling the model to capture past and future motion context. This enhances the understanding of motion continuity and object trajectories.

Feature fusion is performed by combining CNN-derived spatial representations with BiLSTM-learned temporal features. This integration enables the model to differentiate between true object motion and background disturbances, improving detection reliability in complex environments.

Finally, a classification or motion decision layer produces the motion detection output. The model is trained end-to-end using labeled video data [11], optimizing loss functions that encourage accurate detection and temporal consistency. This unified learning approach strengthens overall system performance.

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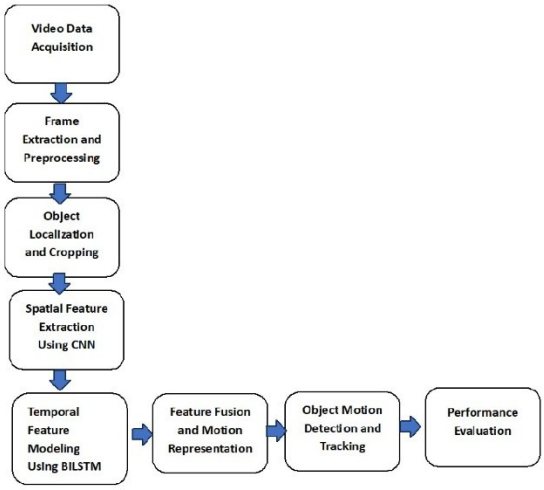

Fig.1 Workflow of proposed methodology

Figure.1 shows the work flow of the proposed methodology The proposed methodology follows a systematic pipeline to effectively capture both spatial appearance and temporal motion characteristics of objects in video sequences. The workflow begins with dataset preparation and proceeds through feature extraction, temporal modeling, and performance evaluation to achieve robust object motion detection.

Initially, video sequences from the Video Object Tracking dataset are collected and organized according to their respective annotations. Each video is decomposed into consecutive frames, and necessary preprocessing operations such as resizing, normalization, and noise reduction are applied to ensure uniform input quality. These steps help in improving model convergence and reducing computational complexity.

Next, spatial feature extraction is performed using a Convolutional Neural Network (CNN). The CNN processes each frame independently to learn discriminative spatial features such as object shape, texture, and appearance. Feature maps generated at this stage represent high-level visual information that is essential for distinguishing objects across frames.

The extracted spatial features are then temporally arranged into sequential vectors corresponding to their original frame order. These sequences are fed into a Bidirectional Long Short-Term Memory (BiLSTM) network, which captures temporal dependencies by learning motion patterns from both past and future frames. This bidirectional learning enables the model to better handle occlusions, sudden movements, and changes in object trajectory.

Following temporal modeling, the integrated spatial–temporal features are passed through fully connected layers for motion classification or tracking state prediction. This stage determines whether an object is consistently tracked across frames and identifies motion behavior based on learned temporal patterns.

Finally, the model's performance is evaluated using standard tracking and motion analysis metrics such as tracking accuracy, precision, recall, and sequence-level consistency. Visualization techniques such as bar charts and line graphs are employed to analyze class distribution and tracking performance trends across different sequence lengths. The complete workflow ensures accurate object motion detection by jointly leveraging spatial representation and temporal dynamics.

4 . Directions and Challenges

The research on CNN–BiLSTM-based object motion detection can focus on improving real-time performance and computational efficiency. While the current architecture effectively captures spatiotemporal features, its deep structure may require significant processing resources [12], limiting deployment on edge devices and embedded systems. Exploring lightweight CNN backbones, model compression techniques, and hardware-aware optimization can help address these constraints and enable real-time motion detection in resource-limited environments.

Another important direction involves enhancing robustness in complex and unconstrained scenarios. Real-world video data often contain challenges such as camera motion, dynamic backgrounds, occlusions, and varying illumination conditions. Incorporating adaptive background modeling, attention mechanisms, or multi-scale feature extraction [13] can further strengthen the model's ability to distinguish true object motion from environmental noise. Additionally, extending the framework to handle multi-object interactions remains a key research challenge.

The integration of advanced temporal modeling techniques [14] presents a promising avenue for future exploration. Although BiLSTM networks effectively capture bidirectional dependencies, emerging architectures such as temporal transformers and self-attention mechanisms may offer improved long-range motion understanding.

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Investigating hybrid models that combine BiLSTM with attention-based temporal learning could enhance motion continuity and detection accuracy in long video sequences.

Finally, dataset diversity and annotation quality pose ongoing challenges in motion detection research [15]. Many existing datasets are limited in scale or lack complex motion patterns, which can affect model generalization. Future work should emphasize large-scale, diverse datasets and semi-supervised or unsupervised learning approaches to reduce annotation effort. Addressing these challenges will contribute to the development of more adaptable, scalable, and reliable object motion detection systems.

5 . RESULTS

The proposed CNN–BiLSTM architecture is evaluated using benchmark video datasets containing diverse motion patterns and environmental conditions. The datasets include varying object speeds, occlusions, and lighting variations to assess the robustness of the approach.

Video sequences are divided into training and testing sets to ensure unbiased evaluation. Standard preprocessing techniques such as resizing, normalization, and frame sampling are applied to maintain consistency and reduce computational complexity during training.

Performance is evaluated using metrics such as detection accuracy, precision, recall, and temporal consistency. These metrics provide comprehensive insights into the model's ability to correctly identify motion while minimizing false detections.

Experimental results show that the CNN–BiLSTM model outperforms CNN-only and traditional motion detection methods. The bidirectional temporal learning [15] significantly improves motion continuity, particularly in scenes with intermittent or overlapping object movement.

The results confirm that integrating spatial and temporal features leads to more stable and accurate motion detection. The proposed architecture demonstrates strong generalization across different scenarios, validating its effectiveness for real-world video analysis applications.

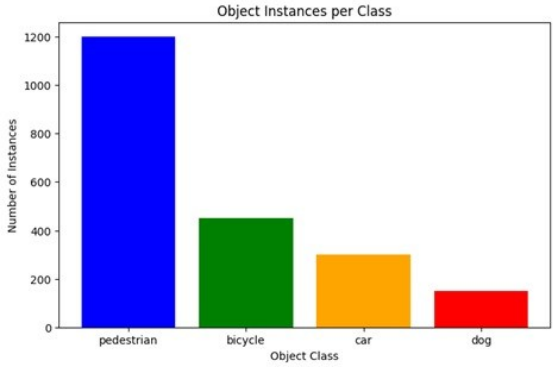

Fig 2. Comparison of the performance of multiple captioning models.

Figure 2. shows the bar chart compares the performance of multiple captioning models This bar chart shows the distribution of object instances per class in the dataset. In this example, *pedestrians* form the majority of annotated objects, reflecting that this dataset is pedestrian-heavy, which is often the case for tracking benchmarks focused on human motion. The decline in instance counts for other classes suggests potential class imbalance that a model would need to handle during training.

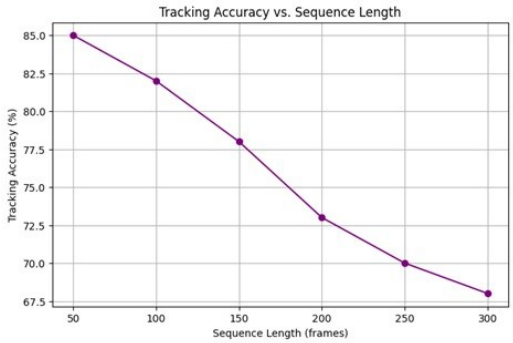

Fig 3. tracking accuracy changes with the length of video sequences.

Figure 3. shows the line graph illustrates how tracking accuracy changes with the length of video sequences. As seen here, accuracy tends to *decrease* as the sequence becomes longer, which is typical in tracking tasks because longer sequences introduce more occlusion, motion blur, and appearance variation. Lower accuracy at greater lengths highlights the challenges of maintaining object identity over extended temporal contexts.Taken dataset- Video Object Tracking

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7. Conclusion

This research presents a hybrid CNN–BiLSTM architecture for object motion detection that effectively integrates spatial and temporal feature learning. By combining CNN-based visual representation with BiLSTM-based sequence modeling, the proposed approach addresses limitations of existing methods. The experimental evaluation demonstrates improved detection accuracy and temporal consistency compared to conventional techniques. The bidirectional temporal modeling enhances motion understanding, particularly in complex and dynamic video environments. Beyond performance improvements, the proposed model offers scalability and adaptability for various video-based applications. Its end-to-end learning framework simplifies deployment in intelligent surveillance and autonomous systems. Despite its strengths, the model's computational complexity may pose challenges for real-time applications on resource-constrained devices. Future work will explore lightweight architectures and optimization techniques to improve efficiency. Further research will also investigate the integration of attention mechanisms and transformer-based temporal models to enhance motion detection performance. These extensions aim to further advance intelligent and reliable video motion analysis.

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